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# Evaluating CNN Architectures and Data Augmentation for Object Recognition

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Seminar: xAI-Sem-??  
Degree: M.Sc. (CitH/AI/...)  
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## Abstract

The abstract paragraph should be indented  $\frac{1}{2}$  inch on both the left- and right-hand margins. Use 10 point type, with a vertical spacing (leading) of 11 points. The word **Abstract** must be centered, bold, and in point size 12. Two line spaces precede the abstract. The abstract must be limited to one paragraph.

## 1 Introduction (E)

Please read the instructions below carefully and follow them faithfully. Those instructions and style file is based on `neurips_2022.tex` (LPPL v1.3c) and was adapted for the purposes at the Chair of Explainable Machine Learning at the University of Bamberg.

## **2 Methodology**

### **2.1 Data Augmentation (E)**

### **2.2 Architectures (E)**

### **2.3 Training (M)**

### **2.4 Metrics (M)**

## **3 Experiments**

## **4 Results**

### **4.1 Results on Testset (D)**

### **4.2 Effects of Data Augmentation (D)**

### **4.3 Class Separability (D)**

## **5 Discussion**

### **5.1 Discoveries (M)**

### **5.2 Limitations (M)**

## **6 Conclusion (E)**

### **6.1 Style**

## **References**

Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to small (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

## **Declaration of Authorship**

Ich erkläre hiermit gemäß § 9 Abs. 12 APO, dass ich die vorstehende Seminararbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe. Des Weiteren erkläre ich, dass die digitale Fassung der gedruckten Ausfertigung der Seminararbeit ausnahmslos in Inhalt und Wortlaut entspricht und zur Kenntnis genommen wurde, dass diese digitale Fassung einer durch Software unterstützten, anonymisierten Prüfung auf Plagiate unterzogen werden kann.

Bamberg, January 27, 2026

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(Ort, Datum)

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(Unterschrift)

## **A Appendix**

Optionally include extra information (complete proofs, additional experiments and plots) in the appendix. This section will often be part of the supplemental material.